Threat-Adaptive Interdiction of Taiwan's Crude Oil Supply

CDT Dennis Pezan
Advisor: LTC Matthew Webb
Department of Mathematical Sciences, United States Military Academy, West Point, NY

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INTRODUCTION

Taiwan imports nearly all of its crude oil by sea, through chokepoints far from home. If China wanted to squeeze Taiwan short of invasion, which disruptions would hurt most?

Applied Problem

- About 794,000 barrels of crude a day, nearly all by sea (EIA, 2025).
- 61% came through the Strait of Hormuz in 2025.
- All three crude ports are on the west coast, facing the mainland.
- About 140 days of crude in storage (Ministry of Economic Affairs, 2026).

Research Questions

- RQ1. With a fixed effort budget, which disruptions reduce Taiwan's crude supply the most?
- RQ2. Can the Threat-Adaptive Network Interdiction Model (TANIM; Webb and Caplice, 2025), built on Ukrainian rail, be applied to sea lanes?

THE NETWORK

The Taiwan Crude Oil Supply Network consists of 61 nodes and 83 edges, from four sources (Gulf, U.S., Oman, West Africa) to Taiwan's three refineries.

- Nodes represent sources, loading terminals, straits and canals, ocean junctions, Taiwan approaches and ports.
- Edges represent supply links, sea routes and port entries. Each edge has a capacity (% of Taiwan's demand it can carry), a cost for China to interdict it, and a chance p that the interdiction succeeds.

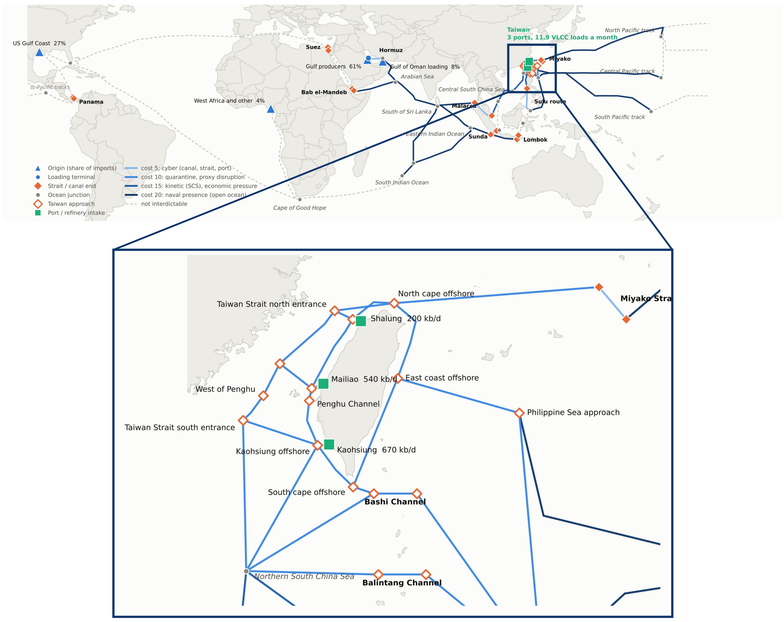


Figure 1. The Taiwan Crude Oil Supply Network (61 nodes, 83 edges). Inset: Taiwan's approaches; all three crude ports face the mainland.

METHODOLOGY

Game Setup

In a Stackelberg (leader-follower) game, one player commits to a move first and the other responds knowing it. China leads by choosing which edges to interdict; Taiwan follows by rerouting its crude to deliver as much as it can (exact max-flow).

- China's budget is fixed at 100. Each interdictable edge has one action with its own cost (5 to 20) and success chance p.
- Score: Taiwan's expected crude delivered, as % of demand. Lower is better for China.

How the RL Agent Plays

- It sees every edge's capacity, cost and p, and the budget left.
- It picks one edge at a time; edges it cannot afford or that carry no flow are masked.
- When the budget is spent, each action succeeds with its p and Taiwan reroutes.

Model and Evaluation

TANIM trains an RL attacker (graph neural network and pointer network, Maskable PPO) on a high-performance computing cluster. It is compared with:

Heuristic	What it is
Random play	Random edges until the budget is spent: the floor
Greedy	Biggest cut per unit of effort, one edge at a time, then swaps
Min-cut	Spends the budget where the network is thinnest
Local search	Simulated annealing over plans within the budget
Optimal (SAA)	Mixed-integer program on sampled outcomes; within 2 points
RL agent	About 3 s per plan; run 4 within 1.4 points of optimal

Table 1. Benchmarks for the RL agent

RESULTS

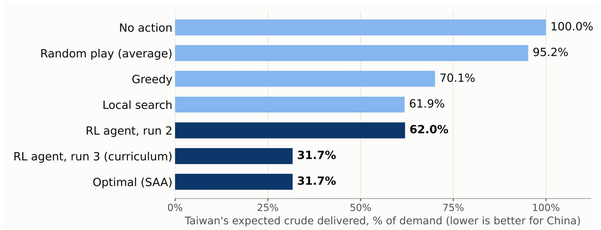


Figure 3. Base case: Taiwan's crude after China spends 100

Taiwan keeps (% of demand); run 4 is trained once, never per scenario:

Scenario	Greedy	Optimal	RL agent
Base case	70.1%	31.7%	31.8%
Costs 50% lower	50.4%	12.9%	12.9%
Costs 50% higher	74.4%	52.8%	52.8%
Taiwan needs 20% more	65.2%	31.7%	31.8%
Taiwan needs 20% less	76.7%	31.7%	31.8%
Hormuz fully shut	49.3%	27.2%	28.6%

Table 2. Overall results (lower is better for China)

WHAT WE FOUND

In run 2 the TANIM attacker settled on a far-away plan: seal the entrances to the South China Sea, pressure the two largest suppliers, and hit the two largest ports. Taiwan keeps 62%, against 70% under greedy. The exact optimum is a ring of ten quarantines around Taiwan's approaches (about 32%). In run 3, half its training episodes started with part of the ring in place; it then built the full ring itself and matched the optimum.

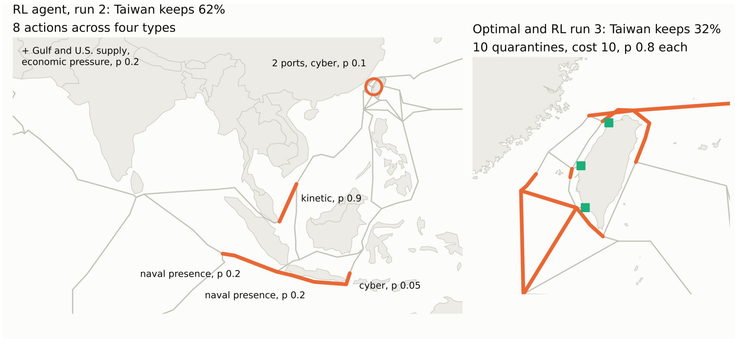


Figure 2. Run 2's plan (left) and the optimal ring, which run 3 also finds (right)

What Was Unexpected

- The best plan is a ring. Each quarantine alone cuts nothing, because Taiwan reroutes, so greedy, local search and run 2 all missed it.
- It is close to Taiwan and high-odds (p 0.8), not a spread of long shots far away.
- Showing the agent what a nearly finished ring is worth was enough: run 3 found the optimum by 300,000 steps.

INSIGHTS

How Might This Generalize?

- Any network where an attacker has a fixed budget and uncertain odds: rail, power, supply chains. Only the edge list changes.

What New Questions Do We Have?

- Trained across budgets (run 4), it is within 1.4 points of optimal in every scenario. Does that hold on new networks?
- How much do the success chances p drive the answer?
- Which edges should Taiwan protect first to break the ring?

References

1. U.S. EIA, Country Analysis Brief: Taiwan, 2026.
2. Webb and Caplice, Solving maximum flow network interdiction using reinforcement learning, SSRN 5392397, 2025.
3. Cormican, Morton and Wood, Stochastic network interdiction, Operations Research, 1998.

ACKNOWLEDGEMENTS

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Keep Clear for Marking